

Causal Inference

Lecture 9

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1 Structural models and G-estimation

- Instrumental variables - Introduction
- Generalized structural mean model

2 Sensitivity analysis

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Basic notations

- Exposure: X , numerical (binary or continuous)
- Outcome: Y , numerical, measured at one specific time point.
- Instrumental variable: Z , numerical (scalar or vector).
- Measured covariates: L , scalar or vector (or an empty set). Possible confounders for X and Y
- Unmeasured confounders: U .

The target parameters

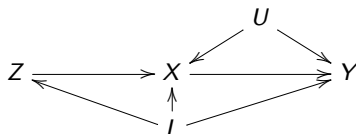
- Our goal is to identify and estimate the causal effect of X on Y , that is denoted by $m^T(L)\psi$, where $m(\cdot)$ is a function of the measured covariates L , and ψ are the causal parameters of interest.

Instrumental variable

- Instrumental variable (IV) regression is a tool that is commonly used in analysis of observational data. Instrumental variables are used to make causal inference about the effect of a certain exposure in a presence of unmeasured confounders.
- A valid instrumental variable is a variable that is associated with the exposure, affects the outcome only through the exposure (**exclusion** criterion), and is unconfounded with the outcome (**exogeneity**).
- The IV assumptions are generally untestable and rely on subject-matter knowledge.
- Therefore, a sensitivity analysis of the G-estimators to invalid instrumental variables is required.
- The one sensitivity parameter method is suitable for linear and non-linear models (specifically, logistic model), and incorporates simultaneously both violations of the **exclusion** and the **exogeneity** assumptions.

Directed acyclic graph (DAG) of a valid instrumental variable

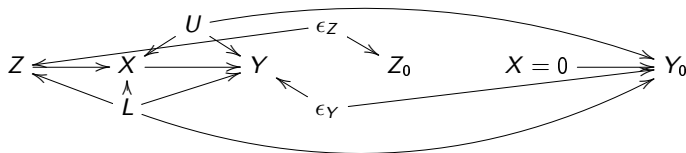
- Let Y be the outcome variable, X the exposure, and Z the instrument. U represents all unmeasured confounders of X and Y , whereas L represents all measured confounders.
- The instrument Z affects Y only through the exposure X , and is unaffected by the unmeasured variables U .



DAG of a causal model with a valid instrumental variable.

Counterfactuals and the twin causal network

- A counterfactual implication of the exogeneity and exclusion assumptions is that a valid IV Z satisfies $Y_0 \perp\!\!\!\perp Z|L$.
- If Z is a valid IV, there is no open path between Y_0 and Z , conditionally on L .



DAG of a twin network causal model with a valid instrumental variable. The left-hand side of the DAG represents the actual world, while the right-hand side represents the hypothetical potential world. $X = 0$ represents the exposure X that is set to 0, and Z_0 represents the potential value of Z in this hypothetical setting.

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GSMM - introduction and notations

- A generalized structural mean model (GSMM) is

$$\xi(\mathbb{E}[Y_1|Z, L, X = x]) - \xi(\mathbb{E}[Y_0|Z, L, X = x]) = m_x^T(L)x\psi_x,$$

where ξ is a link function of a generalized linear model, ψ_x is the vector of causal parameters, and $\dim(m_x(L)) = \dim(\psi_x)$. The composition of $m_x(L)$ defines the exact form of the causal model.

- We exclude U from the condition, since a valid IV allows for identification of the causal parameters.
- The conditioning on the instrument Z is "abuse of notation" as it not simply conditioning, but denote the utilization of Z in order to identify the causal parameters (e.g., TSLS or G-estimation)
- Assume that the association model of the observed outcome follows a parametric structure indexed by the vector of coefficients β_Y , i.e., $\xi(\mathbb{E}[Y|Z, L, X; \beta_Y])$. Such models are referred to as double ξ models.

The coefficients of the observed outcome model

- A generalized structural mean model is

$$\xi(\mathbb{E}[Y_x|L, Z, X = x]) - \xi(\mathbb{E}[Y_0|L, Z, X = x]) = m^T(L)x\psi.$$

- Although the unmeasured confounders are implicit, the consequence of their omission is incorporated in the values of β_Y . Specifically, controlling for the exposure X in the setting with unmeasured confounders U opens a non-causal path between Z and the observed outcome Y and the potential outcome Y_x .
- Formally, X is a collider, thus, conditioning on X opens the non-causal paths

$$Z \rightarrow X \leftarrow U \rightarrow Y \quad \text{and} \quad Z \rightarrow X \leftarrow U \rightarrow Y_x,$$

therefore, Z appears in the GSMM above.

- Since $\xi(\mathbb{E}[Y_x|L, Z, X = x]) = \xi(\mathbb{E}[Y|L, Z, X = x])$ this part is identifiable from the observed model.
- $\xi(\mathbb{E}[Y_0|L, Z, X = x])$ is counterfactual, therefore its identification relies on the availability of a valid IV.

The underlying idea

- A generalized structural mean model is

$$\xi(\mathbb{E}[Y_x|L, Z, X = x]) - \xi(\mathbb{E}[Y_0|L, Z, X = x]) = m^T(L)x\psi.$$

- The underlying idea of the G-estimation approach is to combine the structural and association models to construct a prediction of the counterfactual outcome for each subject.
- Then, this prediction and an the auxiliary function of the instrument Z is used to construct estimating equations.
- The values of the causal parameters that solve these estimating equations are called the G-estimators.
- Loosely speaking, the G-estimators are the values of the causal parameters ψ under which a valid IV assumption $Y_x \perp\!\!\!\perp Z|L$ holds in the observed data.

The general form of $h(\psi)$ for non-linear link-functions

- The G-estimator is the value of the causal parameter ψ under which the assumption $Y_0 \perp\!\!\!\perp Z|L$ holds.
- The potential outcome Y_0 is counterfactual for $x \neq 0$, whose mean is estimated by $h(\psi)$. For a link function ξ , the form of $h(\psi)$ is

$$h(\psi) = \xi^{-1} \left(\xi (\mathbb{E}[Y|X, Z, L]) - m^T(L)X\psi \right).$$

- Let $h(\psi)$ be the residual where the causal effect is removed from the factual outcome Y .
- Special cases of $h(\psi)$ for linear, log, and logit link functions, respectively.

$$h(\psi; \alpha) = \begin{cases} Y - m^T(L)X\psi, \\ Y \exp\{-m^T(L)X\psi\}, \\ \text{expit}(\text{logit}(\mathbb{E}[Y | X = x, Z, L] - m^T(L)X\psi)), \end{cases}$$

Estimating equations for the causal parameter(s) ψ

- $h_i(\hat{\psi})$ as an estimator of the counter-factual value Y_0 , i.e., $\hat{Y}_{0i} = h_i(\hat{\psi})$.
- Since $Y_0 \perp\!\!\!\perp Z|L$ implies that IV assumptions are satisfied, we require $d(L, Z) \perp\!\!\!\perp Y_0|L$. Thus the value of ψ that solves the following estimating equation is the G-estimator of ψ

$$H(\psi) = \sum_{i=1}^n \hat{d}(L_i, Z_i) h_i(\psi) = 0,$$

where $d(L, Z)$ is arbitrary function such that $\mathbb{E}[d(L, Z)|L] = 0$ (more details below).

- The main idea: G-estimator is an estimator which makes the potential outcome Y_0 independent of the instrument Z in the sample, conditionally on the measured covariates L .

The G-estimator

- The consistency of the G-estimator relies on the validity of the instrument Z .
- The violation of the conditional independence of Y_0 and Z can arise either from the **violation of the exclusion**, the **exogeneity assumption**, or both. To illustrate this claim, we use the twin causal network.

Sensitivity parameter α

- We model compositions of mean independence violations using a parametric function $b(L, Z; \alpha)$, such that

$$\xi(\mathbb{E}[Y_0|L, Z]) = a(L) + b(L, Z; \alpha).$$

- In the function $b(L, Z; \alpha)$, the parameter α is a sensitivity parameter that incorporates the conditional association between Y_0 and Z caused by the violation(s). Particularly, we require $b(L, Z; 0) = 0$, and $b(L, Z; \alpha) \neq 0$ for $\alpha \neq 0$.
- For any nonzero value of α , the G-estimator is inconsistent.

Consistency of the G-estimators for invalid IVs

- To ensure the consistency of the G-estimators, we reformulate $h(\psi)$ as a function of α

$$h(\psi; \alpha) = \xi^{-1} \left(\xi(\mathbb{E}[Y|X, Z, L]) - m^T(L)X\psi - b(L, Z; \alpha) \right).$$

- For the true α , α^* , in $h(\psi; \alpha^*)$, the G-estimator of ψ is consistent.
- The true parameter α^* is non-identifiable and in real-world applications is rarely known.
- A sensitivity analysis is carried out by varying α over a range of plausible values, and mapping each value to a corresponding G-estimator.

Linear, log, and logit

$$h(\psi; \alpha) = \begin{cases} Y - m^T(L)x\psi - b(L, Z; \alpha), \\ Y \exp\{-m^T(L)x\psi - b(L, Z; \alpha)\}, \\ \text{expit}(\logit(\mathbb{E}[Y | X = x, Z, L] - m^T(L)x\psi - b(L, Z; \alpha))), \end{cases}$$

$$\mathbb{E}[h(\psi; \alpha) | Z, L] = \mathbb{E}[h(\psi; \alpha) | L] = a(L), \quad \text{for the true value of } \alpha \text{ denoted by } \alpha = \alpha^*$$

- For α^* , the assumption $Y_0 \perp\!\!\!\perp Z | L$ is still violated; however, the estimating functions remain unbiased, and thus the G-estimator is consistent with respect to the true causal parameter ψ .
- The consistency of the G-estimator depends on the correct specification of α in $h(\psi; \alpha)$. Therefore, we define α as a **sensitivity parameter**.
- The exact value of α^* is typically not known to the analyst.
- A **sensitivity analysis** can be carried out by:
 - Varying α over a range of values considered plausible by subject-matter experts.
 - Estimating the causal exposure effect ψ separately for each value of α .

Estimating equations as a function of α

The adjusted estimating equations are:

$$\sum_{i=1}^n Q(Y_i, X_i, L_i, Z_i; \theta, \alpha) = 0$$

- Vector of estimands: $\theta^T = (\beta_Y^T, \mu_Z, \psi)$, and α is the sensitivity parameter.
- Notably, the sensitivity parameter α affects only the estimating equations of ψ , i.e., $d(L, Z; \mu_Z)h(\psi; \alpha)$

$$Q(Y, X, L, Z; \theta, \alpha) = \begin{pmatrix} S(Y, X, L; \beta_Y) \\ S(L, Z; \mu_Z) \\ d(L, Z; \mu_Z)h(\psi; \alpha) \end{pmatrix}$$

Reconstruction of the estimating functions

- Outcome model estimating function $S(Y, X, L; \beta_Y)$:
 - Required when ξ is a non-identity link function.
 - $S(Y, X, L; \beta_Y)$ is the score function of the logistic model.
 - $\frac{1}{n} \sum_{i=1}^n S(Y_i, X_i, L_i; \beta_Y) = 0$ are unbiased estimating functions for the true β_Y i.e., $\mathbb{E}[S(Y, X, L; \beta_Y)] = 0$.
- Instrument model estimating function $S(L, Z; \mu_Z)$:
 - $S(L, Z; \mu_Z)$ is an unbiased estimating function of the instrument model parameters μ .
 - For $L \neq \emptyset$, $S(L, Z; \mu_Z)$ is the score function of the instrument model $\mathbb{E}[Z|L; \mu_Z]$.
- Causal parameters estimating function $d(L, Z; \mu_Z)h(\psi; \alpha)$:
 - $h(\psi; \alpha)$ is the estimator of the counterfactual value Y_0 as defined above that incorporates the sensitivity parameter α .
 - $d(L, Z; \mu_Z)$ is arbitrary function of the the measured covariates L and the instrument Z such that $\mathbb{E}[d(L, Z; \mu_Z)|L] = 0$.
- Note:
 - For the identity link function, no outcome model is needed, thus the estimating functions include only the function of the instrument $S(L, Z; \mu_Z)$ and the causal parameter(s) $d(L, Z; \mu_Z)h(\psi; \alpha)$.

Asymptotic variance

The asymptotic variance of $\hat{\theta}$ is given by the sandwich formula:

$$V(\theta_0, \alpha) = \frac{1}{n} A(\theta_0, \alpha)^{-1} B(\theta_0, \alpha) A(\theta_0, \alpha)^{-T}$$

- Components:

- $A(\theta_0, \alpha) = \mathbb{E} \left[-\frac{\partial Q(\theta_0, \alpha)}{\partial \theta^T} \right]$ is the minus Jacobian of the estimating function $Q(\theta_0, \alpha)$ and known as the “bread” matrix. Where $A(\theta_0, \alpha)^{-1}$ is its inverse, and $A(\theta_0, \alpha)^{-T}$ is the transpose of the inverse matrix.
- $B(\theta_0, \alpha) = \mathbb{E} [Q(\theta_0, \alpha) Q(\theta_0, \alpha)^T]$ is the outer product of the estimating function $Q(\theta_0, \alpha)$ and known as the “meat”.
- θ_0 is the true parameter value, α is the sensitivity parameter, and n is the sample size.

Asymptotic distribution

The vector of estimators is denoted by $\hat{\theta}(\alpha)$. The asymptotic covariance leads to a robust estimator of the coefficients' variance with respect to model misspecification. Asymptotic distribution:

$$\sqrt{n} \left(\hat{\theta}(\alpha) - \theta_0(\alpha) \right) \xrightarrow{D} \mathcal{N}_p(0, V(\theta_0, \alpha))$$

- $\mathcal{N}_p$: Multivariate normal distribution with mean 0 and covariance $V(\theta_0, \alpha)$.
- Subscript p : Dimension of the parametric space $\theta \in \Theta$.
- Superscript D : Convergence in distribution.

Note:

- For the sample version of "bread" and "meat" matrices, replace the expectation operator with the sample average.

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Linear causal model - sensitivity analysis

- Assume that no measured confounders L , so $L = \emptyset$.
- Assume the following structural model:

$$\mathbb{E}[Y_x|Z, U, X = x] - \mathbb{E}[Y_0|Z, U, X = x] = x\psi.$$

- Assume that the violation of valid IV assumptions are specified as:

$$b(Z, L; \alpha^*) = \alpha^* Z.$$

- Derive the populationwise G-estimator of ψ .

Coefficients derivations

Since we have a linear model without measured covariates, we need only the estimating equation for the causal parameter ψ .

1. The adjusted estimating function of the counterfactual Y_0 is
$$h(\psi; \alpha) = Y - \psi X - \alpha^* Z.$$
2. For a linear causal model as in with $L = \emptyset$, the instrument model is

$$D(L, Z; \mu_Z) = S(L, Z; \mu_Z) = Z - \mu_Z,$$

where $\mathbb{E}[Z] = \mu_Z$.

3. We have $m(L) = 1$, thus the unbiased estimating equation for ψ is

$$\mathbb{E} [(Z - \mu_Z)(Y - \psi X - \alpha^* Z)] = 0.$$

Coefficients derivations - solution

3. Solution of the estimating equation

$$\mathbb{E}[(Z - \mu_Z)(Y - \psi X - \alpha^* Z)] = 0,$$

then

$$\begin{aligned}\mathbb{E}[(Z - \mu_Z)(Y - \psi X - \alpha^* Z)] &= \mathbb{E}[Y(Z - \mu_Z) - \psi X(Z - \mu_Z) - \alpha^* Z(Z - \mu_Z)] \\ &= \text{cov}(Y, Z) - \psi \text{cov}(X, Z) - \alpha^* \text{var}(Z) \\ &= 0,\end{aligned}$$

re-arranging the equation, we get

$$\psi = \frac{\text{cov}(Y, Z) - \alpha^* \text{var}(Z)}{\text{cov}(X, Z)} = \frac{\text{cov}(Y, Z)}{\text{cov}(X, Z)} - \frac{\alpha^* \text{var}(Z)}{\text{cov}(X, Z)} = \frac{\beta_{Y.Z}}{\beta_{X.Z}} - \frac{\alpha^*}{\beta_{X.Z}}$$

Correcting the TSLS

- The populationwise true causal effect as a function of α^* :

$$\psi = \frac{\beta_{Y.Z}}{\beta_{X.Z}} - \frac{\alpha^*}{\beta_{X.Z}} = \psi_{TSLS} - \frac{\alpha^*}{\beta_{X.Z}},$$

- Where:

- $\beta_{X.Z} = \text{cov}(X, Z)/\text{var}(Z)$ is the population regression coefficient of the exposure model

$$X = \beta_{0X} + \beta_{X.Z}Z + \epsilon_X.$$

- $\beta_{Y.Z} = \text{cov}(Y, Z)/\text{var}(Z)$ is the population regression coefficient of the outcome-instrument model

$$Y = \beta_{0Y} + \beta_{Y.Z}Z + \epsilon_Y.$$

- Theoretically, if α^* is known, then we can use the TSLS estimator with the aforementioned correction to estimate the causal effect of X on Y .

The explicit form of the sensitivity parameter α^* where $U = \emptyset$

- Assume a linear structural mean model, and that there are no measured and unmeasured confounders, i.e., $U = \emptyset$ and $L = \emptyset$.
- Assume that you use an invalid instrument with the aforementioned violation structure to estimate ψ .
- Recall that the the G-estimator equals the TSLS estimator in linear model. Therefore, the estimator converges to the conditional Wald ratio

$$\begin{aligned}\psi_G &= \frac{\text{cov}(Y, Z)}{\text{cov}(X, Z)} \\ &= \frac{\text{cov}(\beta_0 + \psi X + \beta_{YZ}Z + \epsilon_Y, Z)}{\text{cov}(X, Z)} \\ &= \psi \frac{\text{cov}(X, Z)}{\text{cov}(X, Z)} + \frac{\sigma_Z^2}{\text{cov}(X, Z)} \beta_{YZ} \\ &= \psi + \frac{\beta_{YZ}}{\beta_{X.Z}}.\end{aligned}$$

- In such a case, the sensitivity parameter α^* is the minus of the regression coefficient of Y on Z , $-\beta_{YZ}$, in the outcome model $Y = \beta_0 + \psi X + \beta_{YZ}Z + \epsilon_Y$

Non-identifiable of the true sensitivity parameter - heuristic explanation

Assume, for simplicity, that $L = \emptyset$:

- If $U = \emptyset$, then for the linear model

$$Y = \beta_0 + \psi X + \beta_{YZ.X} Z + \epsilon_Y$$

the sensitivity parameter α^* is

$$\alpha^* = \beta_{YZ.X} = \frac{\text{cov}(Y, Z|X)}{\text{var}(Z|X)}.$$

- Therefore, this parameter can be estimated from the observed data.
- For additive unmeasured confounders U , the true model is

$$Y = \beta_0 + \psi X + \beta_{YZ.XU} Z + \beta_U U + \epsilon_Y$$

$$\alpha^* = \beta_{YZ.XU} = \frac{\text{cov}(Y, Z|X, U)}{\text{var}(Z|X, U)}.$$

- The true sensitivity parameter α^* depends on the covariance of the outcome Y with the IV Z conditional on U (and X), where U is unmeasured, thus non-point identifiable.

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Causal parameters and violation on the logit scale

- Assume a binary outcome Y , a binary exposure X , a binary instrument Z , and an unmeasured confounder U . In addition, assume that there are no measured confounders, i.e., $L = \emptyset$, and $m(L) = 1$. Let ξ be the logit link function. We define the structural mean model as

$$\text{logit}\mathbb{P}(Y_1 = 1|X = 1, Z) - \text{logit}\mathbb{P}(Y_0 = 1|X = 1, Z) = \psi,$$

and the violation as

$$b(Z, L; \alpha^*) = \text{logit}\mathbb{P}(Y_0 = 1|Z) - \text{logit}\mathbb{P}(Y_0 = 1|Z = 0) = \alpha^* Z,$$

where α^* is the true violation parameter.

- Assume also the following logit model for the outcome:

$$\text{logit}(\mathbb{E}[Y|Z, X; \beta_Y]) = \beta_0 + \beta_X X + \beta_Z Z + \beta_{XZ} XZ.$$

- Derive the estimating vector-value function.

- Assuming logistic saturated association “model” for the observed outcome $\mathbb{E}[Y|Z, X; \beta_Y]$, such that $\beta_Y^T = (\beta_0, \beta_X, \beta_Z, \beta_{XZ})$. Therefore, $S(Y, X; \beta_Y)$ are the score functions of this model, particularly

$$S(Y, X; \beta_Y) = \begin{pmatrix} (Y - \mathbb{E}[Y|Z, X; \beta_Y]) \\ (Y - \mathbb{E}[Y|Z, X; \beta_Y])X \\ (Y - \mathbb{E}[Y|Z, X; \beta_Y])Z \\ (Y - \mathbb{E}[Y|Z, X; \beta_Y])XZ \end{pmatrix}.$$

- Since there is no measured confounders, the instrument function is also the score function for the instrument model $d(Z; \mu_Z) = S(Z; \mu_Z) = Z - \mu_Z$.
- The $h(\psi; \alpha)$ is

$$h(\psi; \alpha) = \text{expit}(\text{logit}(\mathbb{E}[Y|Z, X; \beta_Y]) - X\psi - \alpha Z).$$

- Therefore, the G-estimator is the value of ψ that solves

$$\mathbb{E}[(Z - \mu_Z)\text{expit}(\beta_0 + \beta_X X + \beta_Z Z + \beta_{XZ} XZ - X\psi - \alpha Z)] = 0.$$

The system of estimating equations

- Let $\sum_{i=1}^n Q(Y_i, X_i, L_i, Z_i; \theta, \alpha) = 0$ be the estimating equations of the double-logistic causal model, where the vector of estimands is $\theta^T = (\beta_Y^T, \mu_Z, \psi)$, and the estimating vector-valued function

$$Q(Y, X, L, Z; \theta, \alpha) = \begin{pmatrix} S(Y, X, L; \beta_Y) \\ S(L, Z; \mu_Z) \\ d(L, Z; \mu_Z)h(\psi; \alpha) \end{pmatrix}.$$

- Notably, the score function of the instrument model $S(L, Z; \mu_Z)$ equals $d(L, Z; \mu_Z)$.
- This is a system of unbiased estimating equations, i.e., $\mathbb{E}[Q(Y, X, L, Z; \theta, \alpha)] = 0$, therefore, the estimators that solve this system are consistent estimators of the true θ_0 .

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DGP Specification

$$\begin{aligned}Z &\sim \text{Ber}(p_z), \\X|Z = z &\sim \text{Ber}(\text{expit}(\gamma_0 + \gamma_z z)), \\Y|X = x, Z = z &\sim \mathcal{N}(\beta_0 + \beta_x x + \beta_z z + \beta_{xz} xz, \sigma^2).\end{aligned}$$

- Marginal distribution parameters:
 - $P(Z = 1) = p_z = 0.5$.
 - $P(X = 1) = p_x = 0.6$.
- Five out of seven parameters are freely set.

Violation specification

- For $L = \emptyset$ and an invalid IV is modeled as:

$$b(L, Z; \alpha^*) = \alpha^* Z.$$

- Such violation implies:

$$\mathbb{E}[Y_0|Z = 1] - \mathbb{E}[Y_0|Z = 0] = \alpha^*.$$

Expressing $\mathbb{E}[Y_0 | Z]$ in terms of observed data and ψ

The causal model

The assumed latent causal model is

$$\mathbb{E}[Y_0 | X = 1, Z] - \mathbb{E}[Y_1 | X = 1, Z] = \psi.$$

Derivation of $\mathbb{E}[Y_0 | Z]$

We assume that both X and Z are binary.

We use the outcome model, the exposure model, and the causal model to express $\mathbb{E}[Y_0 | Z]$ as a function of the observed data DGP parameters and the causal parameter ψ .

$$\begin{aligned}\mathbb{E}[Y_0 | Z] &= \mathbb{E}_X [\mathbb{E}[Y_0 | Z, X]] \\ &= \mathbb{E}[Y_0 | X = 0, Z] \mathbb{P}(X = 0 | Z) + \mathbb{E}[Y_0 | X = 1, Z] \mathbb{P}(X = 1 | Z) \\ &= (\mathbb{E}[Y | X = 0, Z]) (1 - \mathbb{P}(X = 1 | Z)) + (\mathbb{E}[Y | X = 1, Z] - \psi) \mathbb{P}(X = 1 | Z) \\ &= (\beta_0 + \beta_z Z) (1 - \mathbb{P}(X = 1 | Z)) + (\beta_0 + \beta_x + \beta_z Z + \beta_{xz} Z - \psi) \mathbb{P}(X = 1 | Z) \\ &= (\beta_0 + \beta_z Z) (1 - \text{expit}(\gamma_0 + \gamma_z Z)) + (\beta_0 + \beta_x + \beta_z Z + \beta_{xz} Z - \psi) \text{expit}(\gamma_0 + \gamma_z Z)\end{aligned}$$

Calculating $\mathbb{E}[Y_0 | Z = 1] - \mathbb{E}[Y_0 | Z = 0]$

Using the derived expression for $\mathbb{E}[Y_0 | Z]$, we establish the functional relationship between the sensitivity parameter α^* and the DGP parameters of Y and X .

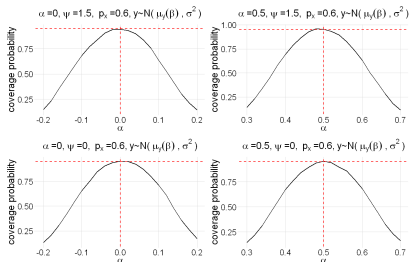
$$\begin{aligned}\mathbb{E}[Y_0 | Z = 1] - \mathbb{E}[Y_0 | Z = 0] &= (\beta_0 + \beta_z)(1 - \text{expit}(\gamma_0 + \gamma_z)) + (\beta_0 + \beta_x + \beta_z + \beta_{xz} - \psi)\text{expit}(\gamma_0 + \gamma_z) \\ &\quad - \beta_0(1 - \text{expit}(\gamma_0)) - (\beta_0 + \beta_x - \psi)\text{expit}(\gamma_0) \\ &= \alpha^*\end{aligned}$$

- This establishes that the difference in expected potential outcomes when setting the exposure X to 0 between $Z = 1$ and $Z = 0$ equals the sensitivity parameter α^* .
- The relationship is derived based on the specified linear causal model, the specified outcome model, and the assumed violation structure.

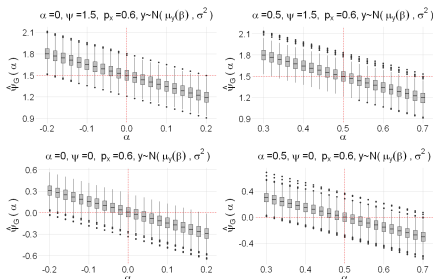
Setting the values of the simulation parameters

- Marginal probabilities:
 - $P(Z = 1) = p_z = 0.5$.
 - $P(X = 1) = p_x = 0.6$.
- Degrees of freedom for the outcome and the instrument model parameters:
 - Three degrees of freedom for $\beta_Y^T = (\beta_0, \beta_x, \beta_z, \beta_{xz})$.
 - One degree of freedom for $\gamma^T = (\gamma_0, \gamma_z)$.
- For the outcome model we set:
 - $\beta_0 = \beta_x = \beta_z = 1$, and solve for β_{xz} .
- For the instrument model we set:
 - $\gamma_0 = -1$, and solve for γ_z .

Linear model example - simulation results



Coverage rates of the 95%-level CIs of the true causal parameter $\psi \in \{0, 0.5\}$ in the logistic structural model as a function of the sensitivity parameter α , for $\alpha^* \in \{0, 0.5\}$, for sample size $n = 1000$, and $m = 1000$ repetitions. $P(Y = 1) = p_y = 0.3$.



Boxplots of empirical distribution of $\hat{\psi}_G(\alpha)$ for the true causal parameter $\psi \in \{0, 0.5\}$ as a function of the sensitivity parameter α , for $\alpha^* \in \{0, 0.5\}$, for sample size $n = 1000$, and $m = 1000$ repetitions. $P(Y = 1) = p_y = 0.3$.

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Logistic model simulation - data generating process (DGP)

- The DGP is

$$Z \sim \text{Ber}(p_z)$$

$$X|Z = z \sim \text{Ber}(\text{expit}(\gamma_0 + \gamma_z z))$$

$$Y|X = x, Z = z \sim \text{Ber}(\text{expit}(\beta_0 + \beta_x x + \beta_z z + \beta_{xz} xz)).$$

- We specify the marginal distribution of Y , X , and Z . In order to relate the violation structure to the DGP parameters of the observed data, we use the fact that a valid IV satisfies $Y_0 \perp\!\!\!\perp Z|L$, particularly, $\mathbb{P}(Y_0|Z = 1) = \mathbb{P}(Y_0|Z = 0)$. For an invalid IV, we define the violation structure as

$$\text{logit}\mathbb{P}(Y_0|Z = 1) - \text{logit}\mathbb{P}(Y_0|Z = 0) = \alpha^*.$$

Expressing $\mathbb{E}[Y_0 \mid Z]$ as a function of observed data and ψ

The causal model

The assumed latent causal model is

$$\text{logit}\mathbb{P}(Y_0|X=1, Z) - \text{logit}\mathbb{P}(Y_1|X=1, Z) = \psi.$$

Double-logit model

Assume that both X and Z are binary.

We use the latent causal model, the the outcome model, and the exposure model to express $\mathbb{E}[Y_0 \mid Z]$ as a function of the observed data DGP parameters and the causal parameter ψ .

$$\begin{aligned}\mathbb{E}[Y_0 \mid Z] &= \mathbb{E}_X [\mathbb{E}[Y_0 \mid Z, X]] \\ &= \mathbb{E}[Y_0 \mid X=0, Z]\mathbb{P}(X=0 \mid Z) + \mathbb{E}[Y_0 \mid X=1, Z]\mathbb{P}(X=1 \mid Z) \\ &= \mathbb{E}[Y \mid X=0, Z]\mathbb{P}(X=0 \mid Z) + \text{expit}(\text{logit}\mathbb{P}(Y_0=1 \mid X=1, Z) - \psi)\mathbb{P}(X=1 \mid Z) \\ &= \mathbb{E}[Y \mid X=0, Z]\mathbb{P}(X=0 \mid Z) + \text{expit}(\text{logit}\mathbb{P}(Y_1=1 \mid X=1, Z) - \psi)\mathbb{P}(X=1 \mid Z) \\ &= \mathbb{E}[Y \mid X=0, Z]\mathbb{P}(X=0 \mid Z) + \text{expit}(\text{logit}\mathbb{P}(Y=1 \mid X=1, Z) - \psi)\mathbb{P}(X=1 \mid Z) \\ &= \text{expit}(\beta_0 + \beta_Z Z)(1 - \text{expit}(\gamma_0 + \gamma_Z Z)) + \text{expit}(\beta_0 + \beta_X + \beta_Z Z + \beta_{XZ} Z - \psi)\text{expit}(\gamma_0 + \gamma_Z Z).\end{aligned}$$

Double-logit model

- The violation on the logit scale is given by

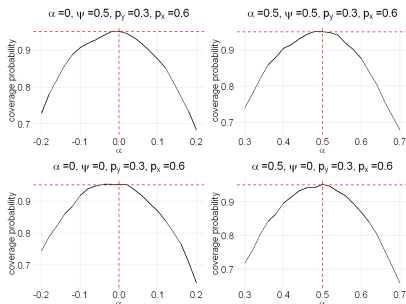
$$\text{logit}\mathbb{P}(Y_0 = 1 \mid Z = 1) - \text{logit}\mathbb{P}(Y_0 = 1 \mid Z = 0) = \alpha^*.$$

- We have nine unknown parameters $\theta_T^0 = (\beta_0, \beta_x, \beta_z, \beta_{xz}, \gamma_0, \gamma_z, \psi, \alpha, p_z)$ and the sensitivity parameter α^* .
- If we specify the causal effect ψ , and the marginal probabilities of $Z = 1$, $X = 1$, and $Y = 1$, we are left with four degrees of freedom.
- Using the result from the previous slide, we have

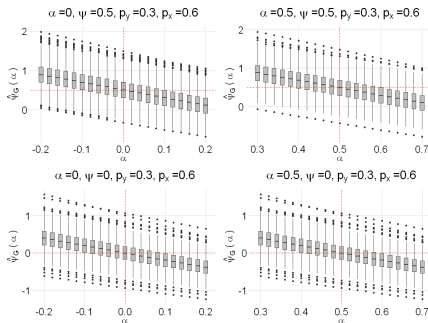
$$\begin{aligned} & \text{logit } \mathbb{P}(Y_0 = 1 \mid Z = 1) - \text{logit } \mathbb{P}(Y_0 = 1 \mid Z = 0) \\ &= \text{logit} \left(\text{expit}(\beta_0 + \beta_z)(1 - \text{expit}(\gamma_0 + \gamma_z)) + \text{expit}(\beta_0 + \beta_x + \beta_z + \beta_{xz} - \psi) \text{expit}(\gamma_0 + \gamma_z) \right) \\ & - \text{logit} \left(\text{expit}(\beta_0)(1 - \text{expit}(\gamma_0)) + \text{expit}(\beta_0 + \beta_x - \psi) \text{expit}(\gamma_0) \right) \\ &= \alpha^*. \end{aligned}$$

- We obtained the functional relationship between the sensitivity parameter α^* and the DGP parameters of the observed outcome Y model, the observed exposure X model, and the latent causal model parameter ψ .

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Simulation overview

- Parameters of the DGP are determined after specifying the true ψ and α^* .
- α^* is used in estimating equations to construct consistent G-estimators of ψ .
- In real-world applications, α^* is rarely known, making plausible interval identification for α challenging.

- Works well for both valid and invalid IVs.
- Applicable to both linear and logistic models.

If $\alpha^* = 0$:

- G-estimator of ψ is consistent for $\alpha = 0$.
- 95%-level CI achieves approximately nominal coverage rate.
- Works well even if the IV is invalid ($\alpha^* = 0.5$).

If $\psi = 0$:

- G-estimator of ψ is consistent for $\alpha = 0.5$.
- 95%-level CI achieves approximately the nominal coverage rate.

Linear Causal Model Sensitivity analysis

- G-estimators are stable with narrow CIs.
- Coverage probability heavily depends on assumed α .
- Deviations of 0.1 from the true α^* (e.g., $\alpha^* = 0$ or $\alpha^* = 0.5$) reduce empirical coverage rates to <70% for nominal 95%-level CIs.
- For correctly specified α :
 - G-estimator is stable.
 - CIs achieve nominal coverage rate.

Logistic Causal Model Challenges

- Variance of G-estimator depends on the dimension and stability of β_Y estimators.
- Compared to linear models the CIs are significantly wider.

Logistic Models

- Highly sensitive to the sample size and extreme values of the marginal probability of Y (e.g., p_Y near 0).
- Low empirical frequencies of combinations of Z, X, Y can result in singular covariance matrices.

Linear Models

- Less sensitive to sample size and extreme values.
- Simpler covariance matrices make G-estimators and CIs more robust.

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One parameter sensitivity analysis - advantages

- The one parameter sensitivity analysis approach theoretical framework and practical guidelines on how to conduct sensitivity analysis of G-estimators using single sensitivity parameter that captures violations of both the exogeneity and the exclusion assumptions.
- Using a single parameter is a valuable advantage as fewer parameters are needed to be specified. This feature may increase model stability and decrease the risk of misspecification.
- The proposed method is applicable both to linear and non-linear causal models.
- The one-sensitivity parameter aligns with and generalize the previous work by Wang et al. (2018).

One parameter sensitivity analysis - disadvantages

- The usage of one parameter for two distinct violations complicates its interpretation.
- Specifying a plausible interval for the sensitivity parameter may be challenging since it relies heavily on subject-matter knowledge and a solid understanding of causal inference.

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References

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